



The period of a non-ideal pendulum

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The period of a pendulum swinging at large angles is dependent on the length of the pendulum and the maximum angle at which it swings. This report seeks to measure the validity of the proposed period equation for pendulums swinging at large angles. It has been observed that longer pendulums yield a higher accuracy in the predicted period.

I. INTRODUCTION

Long ago the pendulum was the world's standard for accurate timekeeping. Pendulum-based clocks kept the world running for centuries [1]. The pendulum is a clever method of utilizing gravitational potential to perform predictable time measurements [2]. In order to measure the time the pendulum takes to swing back and forth, an accurate mathematical description of its motion is necessary.

This report seeks to validate a proposed equation for the period of a real pendulum. A pendulum is a weight suspended from a fixed point that swings back and forth under the influence of gravity. The time it takes for one complete back-and-forth swing, known as a period, is influenced by the length of the pendulum, the angle at which it swings, and the acceleration due to gravity.

The equation for the period of an ideal pendulum swinging at small angles is a well-defined, closed-form solution that accurately predicts pendulum motion. However, the period of a pendulum swinging at large angles requires the use of an elliptic integral, which must be evaluated numerically by a computer [3]. In order to prove the accuracy of the proposed solution, experimental measurements were taken of the period of varying pendulum lengths and angles of oscillation.

II. FINDINGS

A. Mathematical Background

Under the acceleration due to gravity, g , there exists a restoring torque

$$\tau = mgL \sin \theta, \quad (1)$$

that causes angular acceleration on a mass m with moment of inertia

$$I = mL^2, \quad (2)$$

where L is the distance from the point of oscillation and the center of mass of the pendulum. Newton's 2nd law

for this scenario then becomes

$$mgL \sin \theta = mL^2 \frac{d^2\theta}{dt^2}, \quad (3)$$

which simplifies down to the angular acceleration for a pendulum given by

$$\frac{d^2\theta}{dt^2} = -\frac{g}{L} \sin \theta, \quad (4)$$

where θ is the angle between the pendulum and the direction of gravitational force (the center-line) and t is time. If the pendulum is swinging at small angles, where $\sin(\theta) \approx \theta$ is a good approximation, the period of the pendulum is a function of length given by

$$T_0(L) = 2\pi \sqrt{\frac{L}{g}}. \quad (5)$$

If the pendulum is swinging at large angles, eq. (4) becomes more difficult to solve. The solution requires the use of an elliptic integral and is given by

$$T(L, \theta_{\max}) = T_0(L) \frac{2}{\pi} K \left(\sin \left(\frac{\theta_{\max}}{2} \right) \right), \quad (6)$$

where T_0 is the small angle period result in eq. (5) and $K(x)$ is the complete elliptic integral of the first kind defined as

$$K(x) = \int_0^{\pi/2} \frac{1}{\sqrt{1 - x^2 \sin^2 u}} du. \quad (7)$$

B. Analysis

The period was recorded for pendulums of varying lengths ranging from 7.4 to 31.6 cm and maximum oscillation angles ranging from 0.1 to $\frac{\pi}{2}$ radians. The predicted and measured periods as a function of maximum angle are shown in fig. 1 with colors identifying the various lengths. In the case of a long pendulum, the predicted period of oscillation is an accurate description for

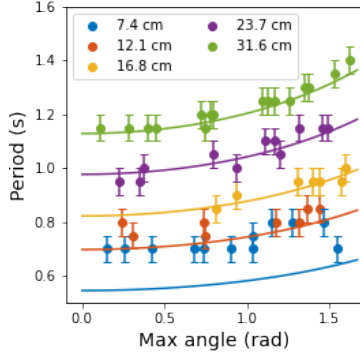


FIG. 1. Predicted and measured values of pendulum period for varying lengths. Lines represent predicted values, scatter points are measured values, and each color indicates a different length pendulum.

Pendulum Length (cm)	R^2
7.4 ± 0.2	0.55
12.1 ± 0.2	0.77
16.8 ± 0.2	0.91
23.7 ± 0.2	0.93
31.6 ± 0.2	0.98

TABLE I. Correlation coefficients.

the entire range of angles. The predicted period of short pendulums is less fitting throughout the entire range of angles. The correlation coefficient for each pendulum length is shown in table I and represents how well the predicted periods match the measured periods. There exists a relationship between the length of the pendulum and the strength of correlation between the theoretical and measured periods.

III. CONCLUSION

For all recorded lengths, the period of a real pendulum swinging at a large angles oscillate with a period that is larger than when the angle is very small. Thus, the small angle approximation will not predict the period of a pendulum for all oscillation angles. The period of a pendulum swinging at large angles as predicted by eq. (6) is accurate for pendulums of relatively long length; when the added masses are below the halfway point of the length of the rod being used.

The source of the inaccuracy of the period prediction equation at short pendulum lengths may be due to the underlying mathematical assumptions. The mathemati-

cal background starting with eq. (2) is set up under the assumption that the pendulum is a point mass held at a distance L by a mass-less rod. The length measurements are representative of the distance between the oscillation point and the center of mass of the pendulum as calculated by the experimenters. The assumption is that we can treat the pendulum as the entire mass being a point mass at the center of mass distance, therefore simplifying the moment of inertia calculation which was used to set up eq. (4). For situations when the added masses are at the far end of the rod, this is a good approximation. However, when the masses are put up at the top of the rod, this becomes a poor estimation, as the mass of the rod below is influencing the motion. Future work could be performed to restructure the mathematics to include a more accurate moment of inertia.

Appendix A: Uncertainty analysis

The uncertainty of measurements in length (δ_L) and period (δ_T) are as follows

$$\delta_L = 0.2 \text{ cm} \quad (\text{A1})$$

$$\delta_T = 0.05 \text{ s} \quad (\text{A2})$$

The uncertainty in length is determined by two of the smallest divisions on the ruler used to measure the distance between the mass and the pivot point. The uncertainty in period for the pendulum was determined by the sampling frequency of the Pasco software, 20 Hz which equates to one sample every 0.05 seconds.

The center of mass distance from the oscillation point of the pendulum is given as follows

$$L_{CM} = \frac{L_{rod}M_{rod} + L_{mass}M_{mass}}{M_{rod} + M_{mass}}, \quad (\text{A3})$$

where L_{rod} and M_{rod} are the effective length and mass of the rod, and L_{mass} and M_{mass} are the distance from the oscillation point and mass of the added masses to the rod.

In the case of our experiment,

$$M_{rod} = 22.8g, \quad (\text{A4})$$

$$M_{mass} = 150.8g. \quad (\text{A5})$$

The uncertainty in center of mass distance calculations given eq. (A1) are

$$\delta_{L_{cm}} = \left| \frac{M_{mass}}{M_{rod} + M_{mass}} \delta_L \right| \quad (\text{A6})$$

$$\approx 0.2 \text{ cm}. \quad (\text{A7})$$

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